COUNTABLE FAMILY OF RELATIVE EQUIVALENCES CATCHING ALL FIBRES

THE STACKS PROJECT AUTHORS

Editorial note. Relies on same-chapter proposition-equivalence for the representation step; its candidate proof is a 132-byte stub and it is not carded; transcribed as-is.

Allowed standard prerequisites (editorial).

- Countable classification. lemma-countable-finite-type (same chapter, candidate proof is a 32-byte stub, not carded). Countably many isomorphism classes of systems over the countable subfield.
- Countable perfect categories. lemma-countable-perfect (same chapter, judged citation-only, not carded). Second input of the countability pair.
- Base change and descent of kernels. lemma-base-change-rek L26476, lemma-descend-rek L26473 (carded this batch). Base change preserves relative equivalences; kernels descend the limit tower.

Source reference. Original lemma 0G0K: sources/stacks-equiv/pinned-original-source.tex, source lines 3838–3838. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0G0U: sources/stacks-equiv/pinned-original-source.tex, source lines 4309–4309. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0G0W: sources/stacks-equiv/pinned-original-source.tex, source lines 4343–4343. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0G0L: sources/stacks-equiv/pinned-original-source.tex, source lines 3888–3888. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0FYY: sources/stacks-equiv/pinned-original-source.tex, source lines 1645–1645. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is

Date: stacks-project:equiv.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0G0E: sources/stacks-equiv/pinned-original-source.tex, source lines 3569–3569. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01ZM: focused-seeds/geometry/S9-eqv-countable-equivs/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0C0C: focused-seeds/geometry/S9-eqv-countable-equivs/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 081F: focused-seeds/geometry/S9-eqv-countable-equivs/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original proposition 0G0D: sources/stacks-equiv/pinned-original-source.tex, source lines 3553–3553. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let K be an algebraically closed field. There exists a countable set I and for $i \in I$ a pair $(S_i/K, X_i \to S_i, Y_i \to S_i, M_i)$ with the following properties*

1. *S_i is a scheme of finite type over K,*
2. *$X_i \to S_i$ and $Y_i \to S_i$ are proper smooth morphisms of schemes,*
3. *$M_i \in D_{perf}(\mathcal{O}_{X_i \times_{S_i} Y_i})$ is the Fourier-Mukai kernel of a relative equivalence from X_i to Y_i over S_i, and*
4. *for any smooth proper schemes X and Y over K such that there is a K-linear exact equivalence $D_{perf}(\mathcal{O}_X) \to D_{perf}(\mathcal{O}_Y)$ there exists an $i \in I$ and a $s \in S_i(K)$ such that $X \cong (X_i)_s$ and $Y \cong (Y_i)_s$.*

Proof. Choose a countable subfield $k \subset K$ for example the prime field. By Lemmas 0G0U and 0G0W there exists a countable set of isomorphism classes of systems over k satisfying parts (1), (2), (3) of the lemma. Thus we can choose a countable set I and for each $i \in I$ such a system

$$(S_{0,i}/k, X_{0,i} \to S_{0,i}, Y_{0,i} \to S_{0,i}, M_{0,i})$$

over k such that each isomorphism class occurs at least once. Denote $(S_i/K, X_i \to S_i, Y_i \to S_i, M_i)$ the base change of the displayed system to K. This system has properties (1), (2), (3), see Lemma 0G0K. Let us prove property (4).

Consider smooth proper schemes X and Y over K such that there is a K-linear exact equivalence $F : D_{perf}(\mathcal{O}_X) \to D_{perf}(\mathcal{O}_Y)$. By Proposition 0G0D we may assume that there exists an object $M \in D_{perf}(\mathcal{O}_{X\times Y})$ such that $F = \Phi_M$ is the corresponding Fourier-Mukai functor. By Lemma 0FYY there is an M' in $D_{perf}(\mathcal{O}_{Y\times X})$ such that $\Phi_{M'}$ is the right adjoint to Φ_M. Since Φ_M is an equivalence, this means that $\Phi_{M'}$ is the quasi-inverse to Φ_M. By Lemma 0FYY we see that the Fourier-Mukai functors defined by the objects

$$A = R\mathrm{pr}_{13,*}(L\mathrm{pr}_{12}^*M \otimes^{\mathbf{L}}_{\mathcal{O}_{X\times Y\times X}} L\mathrm{pr}_{23}^*M')$$

in $D_{perf}(\mathcal{O}_{X\times X})$ and

$$B = R\mathrm{pr}_{13,*}(L\mathrm{pr}_{12}^*M' \otimes^{\mathbf{L}}_{\mathcal{O}_{Y\times X\times Y}} L\mathrm{pr}_{23}^*M)$$

in $D_{perf}(\mathcal{O}_{Y\times Y})$ are isomorphic to $\mathrm{id} : D_{perf}(\mathcal{O}_X) \to D_{perf}(\mathcal{O}_X)$ and $\mathrm{id} : D_{perf}(\mathcal{O}_Y) \to D_{perf}(\mathcal{O}_Y)$ Hence $A \cong \Delta_{X/K,*}\mathcal{O}_X$ and $B \cong \Delta_{Y/K,*}\mathcal{O}_Y$ by Lemma 0G0E. Hence we see that M is the Fourier-Mukai kernel of a relative equivalence from X to Y over K by definition.

We can write K as the filtered colimit of its finite type k-subalgebras $A \subset K$. By Limits, Lemma 01ZM we can find X_0, Y_0 of finite type over A whose base changes to K produces X and Y. By Limits, Lemmas 081F and 0C0C after enlarging A we may assume X_0 and Y_0 are smooth and proper over A. By Lemma 0G0L after enlarging A we may assume M is the pullback of some $M_0 \in D_{perf}(\mathcal{O}_{X_0\times_{\mathrm{Spec}(A)}Y_0})$ which is the Fourier-Mukai kernel of a relative equivalence from X_0 to Y_0 over $\mathrm{Spec}(A)$. Thus we see that $(S_0/k, X_0 \to S_0, Y_0 \to S_0, M_0)$ is isomorphic to $(S_{0,i}/k, X_{0,i} \to S_{0,i}, Y_{0,i} \to S_{0,i}, M_{0,i})$ for some $i \in I$. Since $S_i = S_{0,i} \times_{\mathrm{Spec}(k)} \mathrm{Spec}(K)$ we conclude that (4) is true with $s : \mathrm{Spec}(K) \to S_i$ induced by the morphism $\mathrm{Spec}(K) \to \mathrm{Spec}(A) \cong S_{0,i}$ we get from $A \subset K$. □